Vishwa Patel

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Education

University of Southern California, M.S. in Computer Science, **GPA:** 3.66/4.0 — Los Angeles, CA
Jan 2026 – Dec 2027

- **Coursework:** Analysis of Algorithms, Database Systems, Web Technologies, NLP, Agentic AI
- Serve as a **Grader** for **280** students in USC CSCI 585 Database Systems, designing exam questions and scoring rubrics
- Guided **1,200+** incoming master's students as an **Orientation Leader** at USC Viterbi with **20+** fellow leaders

Gujarat Technological University, B.E. in Computer Engineering, **GPA:** 3.95/4.0 — Gujarat, India
Aug 2019 – Jun 2023

- **Coursework:** Data Structures, Operating Systems, Software Engineering, Machine Learning, AI

Technical Skills

Programming Languages: Python, Java, C#, C++, JavaScript, TypeScript, HTML, CSS, C, SQL, Go

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, pgvector, Query Optimization

Full Stack: FastAPI, Spring Boot, .NET MVC, React, REST APIs, Microservices, RabbitMQ

AI & ML: AI Agents, LangChain, LangGraph, RAG, NLP, PyTorch, scikit-learn, Gemini, Codex, Copilot, Claude Code, MCP

Cloud, DevOps & Testing: AWS, Docker, Kubernetes, CI/CD, Git, GitHub, Linux, Pytest, JUnit, Postman

Professional Experience

Software Engineer, Casepoint Pvt. Ltd – Gujarat, India — **Jul 2023 – Jun 2025**

- Increased document-processing throughput by **90%** by diagnosing a file-locking constraint that serialized file handling, re-architecting the **C#/.NET** system to run in parallel, and releasing it through **CI/CD**
- Recovered **70%** of failed **PST** files by creating a repair service that detects failures in the processing pipeline, lets clients trigger reprocessing, and automatically routes repaired files back into it
- Reduced file errors by **40%** by upgrading third-party libraries, including **Oracle**, **Aspose**, and **VintaSoft**
- Processed **100%** of **10K+** previously skipped forensic files by building a full-stack feature integrated into the pipeline
- Cut status-update latency by **80%** by buffering each service's updates in **Redis** and batching **PostgreSQL** commits
- Earned the **2024 Excellence Award** for sprint delivery by shipping **10+** features in one quarter with **Agile Scrum**
- Identified and optimized **85%** of slow-running queries across processing services using **Elasticsearch** query monitoring
- Enabled **90%** of clients to customize reports with only required fields, replacing fixed unnecessary ones, by supporting **20+** processing-team database fields and coordinating with **5+** teams to prevent crashes

Software Engineer Intern, Casepoint Pvt. Ltd – Gujarat, India — **Jan 2023 – Jun 2023**

- Saved HR **60%** of hiring time by building a Recruitment System with **20+** UI screens centralizing scheduling, scored reports, resume review, offer letters, and tracking in **.NET MVC**, **C#**, **PostgreSQL**, and **Kendo UI**
- Lowered average response time by **35%** by integrating **RabbitMQ** for asynchronous messaging between app components

Software Engineering Intern, Amiseq – Pune, India — **Jun 2022 – Jul 2022**

- Delivered an internal to-do list backend with **10+ REST API** endpoints by building modular **Java** and **Spring Boot** microservices on **PostgreSQL**, reaching **95%** test coverage with **JUnit** and validating with **Postman**

Projects

MCP-PROBE: Do MCP Tool Poisoning Detectors Generalize? | Python, PyTorch — **Oct 2026** | **GitHub**

- Built a Python proxy between AI agents and MCP servers blocking tools flagged by MCP-Guard and MCP Guardian, fine-tuning MCP-Guard variants with **3** MCPTox categories held out for **99.9%** accuracy vs. rule-based Guardian's **50.2%**

Haggle: Multi-Agent Event Catering Platform | Python, Fetch.ai uAgents, ASI:One, Stripe — **Oct 2026** | **GitHub**

- Implemented a **6**-agent system in Python for event catering where agents find nearby caterers, collect quotes, negotiate up to **3** rounds within each vendor's rules, and book in one chat, cutting manual steps by **63%**

TechHire: AI Job Search Platform | FastAPI, React, PostgreSQL, pgvector, LangGraph — **Aug 2026** | **GitHub**

- Developed an AI-powered job search platform with FastAPI, React, and PostgreSQL that lists **1,000+** postings with sponsorship, salary, and skill filters, plus a resume scorer and **3**-model job summaries
- Designed a LangGraph RAG chatbot on pgvector that answers questions with cited sources, lifting search accuracy from **68%** to **92%** on **74** test questions, and cut repeat-search time **93%** with Redis caching